

Problem 1 : Basic Exam, Spring 2022

Let (a_n) be a sequence of positive reals such that

$$a = \lim_{n \rightarrow \infty} na_n$$

exists with $a \in (0, \infty)$. Prove that $f(x) = \sum_{n=0}^{\infty} a_n x^n$ is convergent with $f(x)$ continuously differentiable for all $x \in (-1, 1)$. Then show

$$\lim_{x \rightarrow 1^-} f'(x)(1-x) = a.$$

Solution.

Problem 2 : Basic Exam, Fall 2021

Let $\{a_n\}$ be a sequence in $\mathbb{R}$ that decreases to 0 and let $\{b_n\}$ be a sequence with bounded partial sums, i.e. there exists $M \in \mathbb{R}$ such that for all n ,

$$\left| \sum_{k=1}^n b_k \right| \leq M.$$

Show that $\sum_{n=1}^{\infty} a_n b_n$ converges.

Solution. Let

$$B_n = \sum_{k=1}^n b_k \text{ and } B_0 = 0.$$

Thus

$$\begin{aligned} S_n &= \sum_{k=1}^n a_k b_k = \sum_{k=1}^n a_k (B_k - B_{k-1}) \\ &= \sum_{k=1}^n a_k B_k - \sum_{k=1}^n a_k B_{k-1} \\ &= a_n B_n + \sum_{k=1}^{n-1} a_k B_k - \sum_{k=0}^{n-1} a_{k+1} B_k \\ &= a_n B_n + \sum_{k=1}^{n-1} (a_k - a_{k+1}) B_k \end{aligned}$$

$\forall \varepsilon > 0$. Since $a_n \rightarrow 0$, $\exists N_1 \in \mathbb{N}$ such that $\forall n \geq N_1: |a_n| < \frac{\varepsilon}{3M}$. Furthermore, $\exists N_2 \in \mathbb{N}$ such that $\forall m \geq n \geq N_2$ we have that $|a_m - a_n| < \frac{\varepsilon}{3M}$. Thus $\forall m \geq n \geq \max(N_1, N_2)$:

$$\begin{aligned}
|S_m - S_n| &= \left| a_m B_m + \sum_{k=1}^{m-1} (a_k - a_{k+1}) B_k - a_n b_n - \sum_{k=1}^{n-1} (a_k - a_{k+1}) B_k \right| \\
&\leq M|a_m| + M|a_n| + \left| \sum_{k=n}^{m-1} (a_k - a_{k+1}) B_k \right| \\
&\leq M|a_m| + M|a_n| + M \left| \sum_{k=n}^{m-1} (a_k - a_{k+1}) \right| \\
&= M|a_m| + M|a_n| + M|a_n - a_m| < \varepsilon
\end{aligned}$$

Thus S_n is Cauchy hence S_n is convergent.

Problem 3 : Basic Exam, Spring 2018

Prove that for each $p \in \mathbb{N}$, the infinite series

$$\sum_{n=1}^{\infty} \frac{\sin(\pi n/p)}{n}$$

converges.

Solution.

Problem 4

Assume that a positive series $\sum_{n=1}^{\infty} a_n$ converges and that the sequence $(na_n)_{n=1}^{\infty}$ is decreasing. Show that

$$\lim_{n \rightarrow \infty} (n \log n) a_n = 0.$$

Solution.